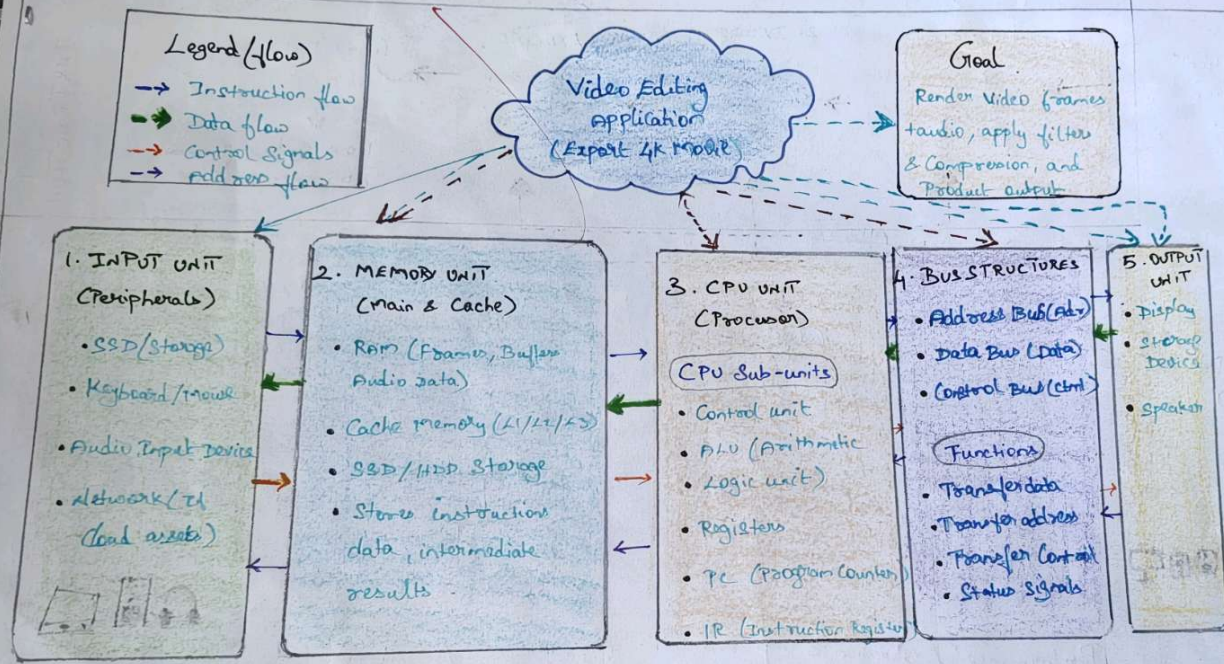


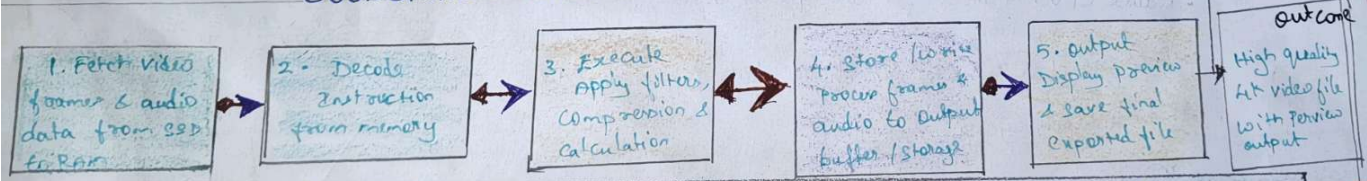
15/11/2024

Computer Architecture / CS AI 214
 Assignment Tool - 2 / Concept Map
 Col - AT2

1) VIDEO RENDERING: COORDINATION OF FIVE FUNCTIONAL UNITS



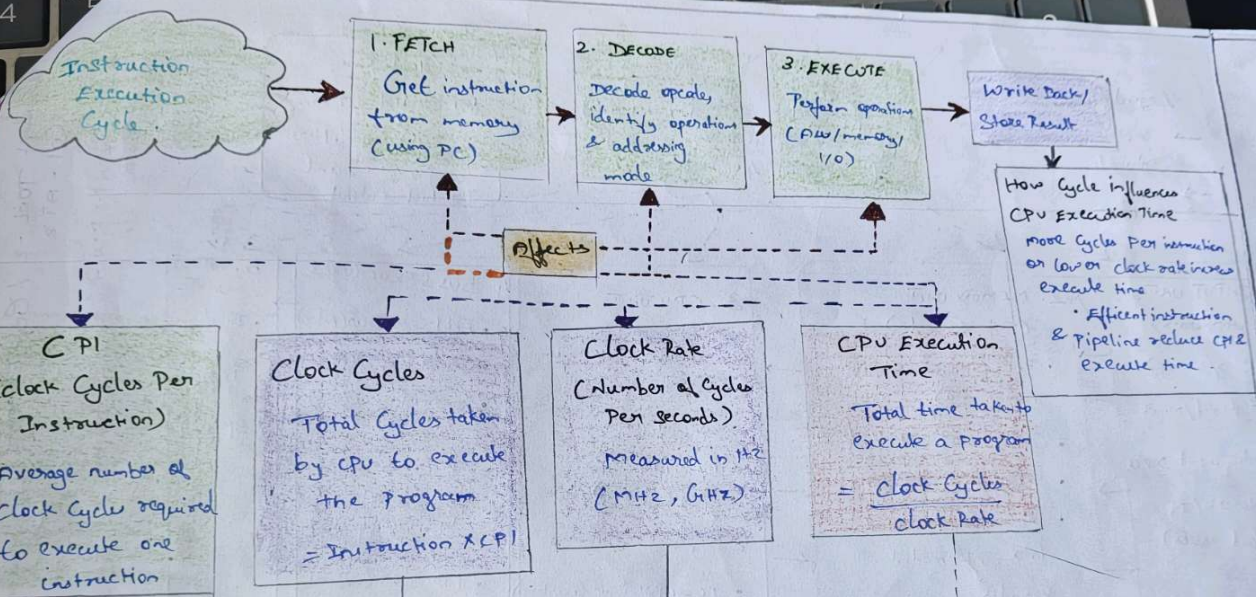
COORDINATION WORKFLOW DURING RENDERING



The five functional units works together through the bus system, The CPU operations, memory storage data/instruction, I/O device provide input and output and buses enable communication.

B. Sivarangani
 192521575
 B. Tech IT

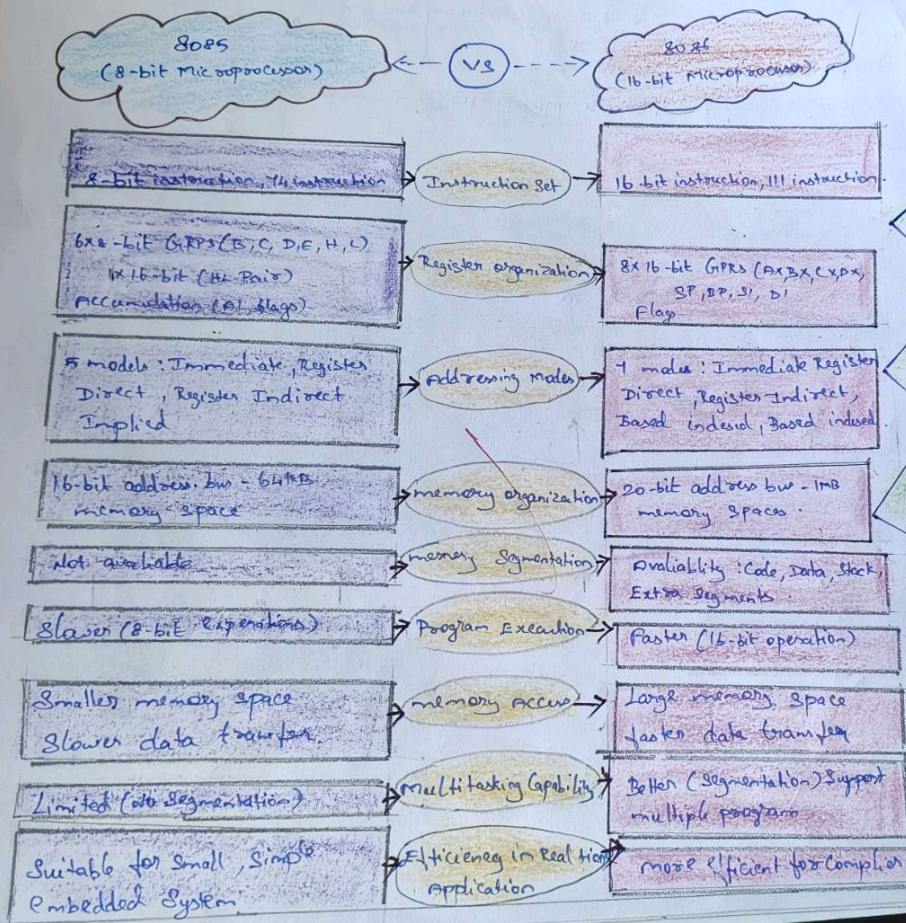
2) INSTRUCTION EXECUTION CYCLE & PERFORMANCE PARAMETER



- Performance Evaluation & improvement**
- Reduce CPU: optimize code, use efficient instructions, Pipelining, Caching
 - Increase clock rate: better hardware design (up to thermal/Power limits)
 - Reduce clock Cycles: algorithm optimization, Parallel Processing
 - Minimize Execution Time: Faster program Performance.

Key Relationship

$$\text{CPU Execution Time} = \frac{\text{Instruction} \times \text{CPI}}{\text{Clock Rate}}$$



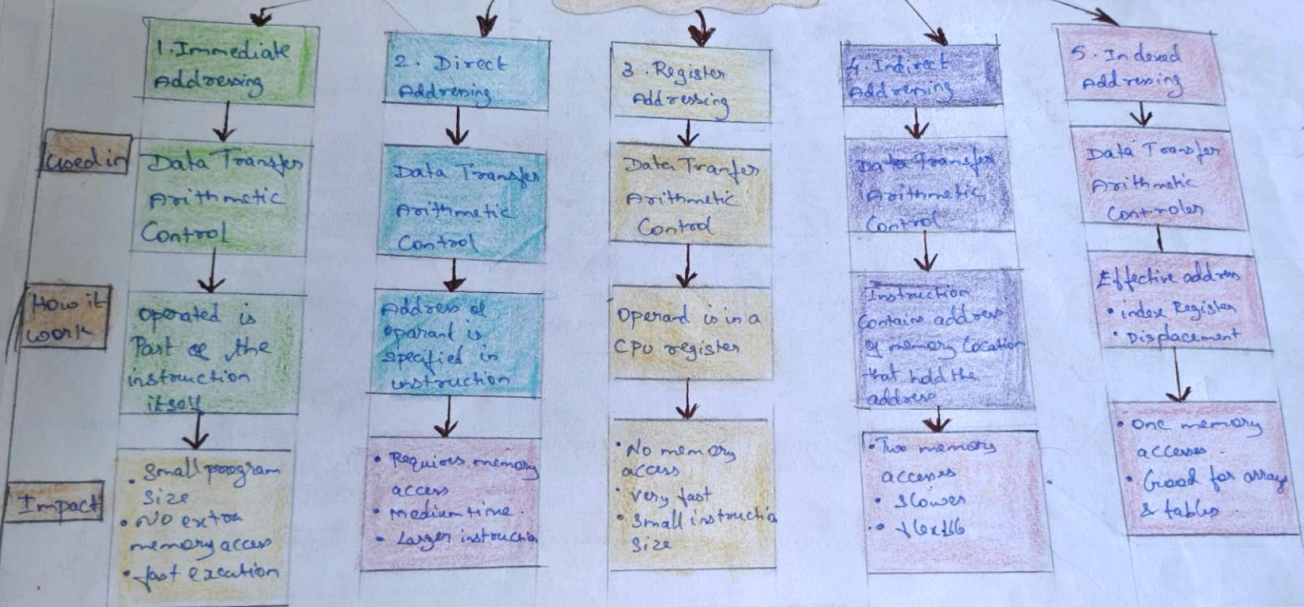
Impact of Architectural Features

- 16-bit data path & larger address space in 8086 → faster execution.
- more addressing modes in 8086 → flexible memory access
- Segmentation in 8086 → better program organization & Multitasking
- Better memory management in 8086 → Higher efficiency in complex application.

8085 vs 8086 ARCHITECTURAL COMPARISON

4) ADDRESSING MODES & THEIR IMPACT

Addressing modes



Real-world applications

- Constants
- Initialization
- Eg. `mov A, #25H`

- Global variable
- I/O mapped data
- Eg. `mov A, 2050H`

- Counters
- Temporary data
- Eg. `ADD B, C`

- Linked lists
- Pointers
- Eg. `mov A, @R0`

- Array elements
- Table Lookup
- Eg. `mov A, TAB`

`R0` → `Cache`

`SI` → `Cache`

